

Roll No.

--	--	--	--	--	--	--	--

END SEMESTER SUPPLEMENTARY EXAMINATION 2024

Name of the Course: **B.Tech**

Name of the Paper –Computer Networks - II

Semester: **VII**Paper Code: **TCS703**Time: **3 Hours**Maximum Marks: **100****Note:-**

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	For the network provided in the figure with given link costs, apply Dijkstra's shortest-path algorithm as used in link-state routing to compute the shortest paths from node D to all other network nodes.	CO5/ CO6
(b)	(i) What are the key differences between global routing and decentralized routing? (ii) How does broadcast routing differ from multicast routing?	
(c)	Consider a network topology with four routers A, B, C, and D, where their initial distance vectors are as follows: - Router A: [0, 3, 7, ∞] - Router B: [3, 0, ∞, 2] - Router C: [7, ∞, 0, ∞] - Router D: [∞, 2, ∞, 0] The cost from Router A to Router B is 3, Router B to Router D is 2, and Router D to Router A is 4. i) Using the Bellman-Ford equation, update the distance vectors for each router after one iteration based on the above information. ii) Show the convergence of the Distance Vector Algorithm by indicating the number of iterations required until no further changes occur in the distance vectors.	
Q2	(20 marks)	
(a)	Explain in detail the functionalities, characteristics, and differences of the BGP, OSPF, and RIP routing protocols?	
(b)	Explain the concept of parity checking as an error detection technique in data transmission. Describe both even parity and odd parity methods, highlighting how they detect errors in transmitted data.	CO2/ CO3

(c)	What are the challenges and limitations of CSMA and its variants in managing network congestion, reducing latency, and ensuring fairness among nodes competing for medium access?	
Q3	(20 marks)	
(a)	i) Pure Aloha and Slotted Aloha. ii) Discuss services offered by data link layer to network layer	CO3/ CO2
(b)	How does RTSP differ from HTTP, and what specific functionalities and advantages make RTSP more suitable for streaming media delivery?	
(c)	What is the Session Initiation Protocol (SIP), and why is it important for real-time communication? Describe the core functions of SIP in setting up and managing multimedia sessions.	
Q4	(20 marks)	
(a)	What is the Reverse Address Resolution Protocol (RARP), and what role does it play in computer networking? How does RARP function, and how does it differ from ARP in terms of operation and use cases?	CO4/ CO3
(b)	How SNMP assist in Network Management Explain the structure of SNMP messages, including the various types of messages (GET, GETNEXT, SET, TRAP) used in SNMP operations.	
(c)	Define H.323 and describe its primary components. Explain the roles of these components such as terminals, gateways, gatekeepers, and multipoint control units (MCUs) in the H.323 architecture.	
Q5	(20 marks)	
(a)	What is the Transmission Control Protocol (TCP), and how does it ensure reliable data transmission? Additionally, what is an Echo Server, and how does it function within a network environment?	CO 5/CO6
(b)	Discuss Secure socket layer and Token passing protocol	
(c)	The TCP client and server follow the flow of functions that we diagrammed in the given figure. Write a code (any language) for Concurrent server <pre> graph LR stdin -- fgets --> TCP_client[TCP client] TCP_client -- fputs --> stdout TCP_client -- write --> TCP_server[TCP server] TCP_server -- read --> TCP_client TCP_server -- write --> TCP_client </pre>	